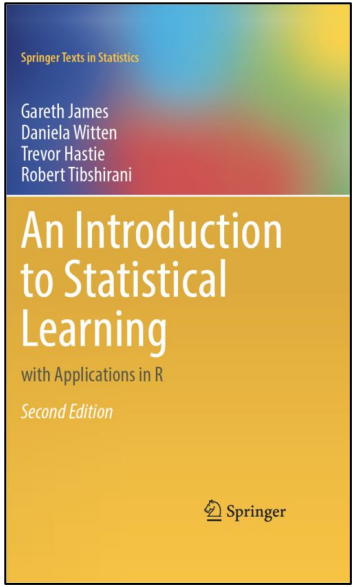
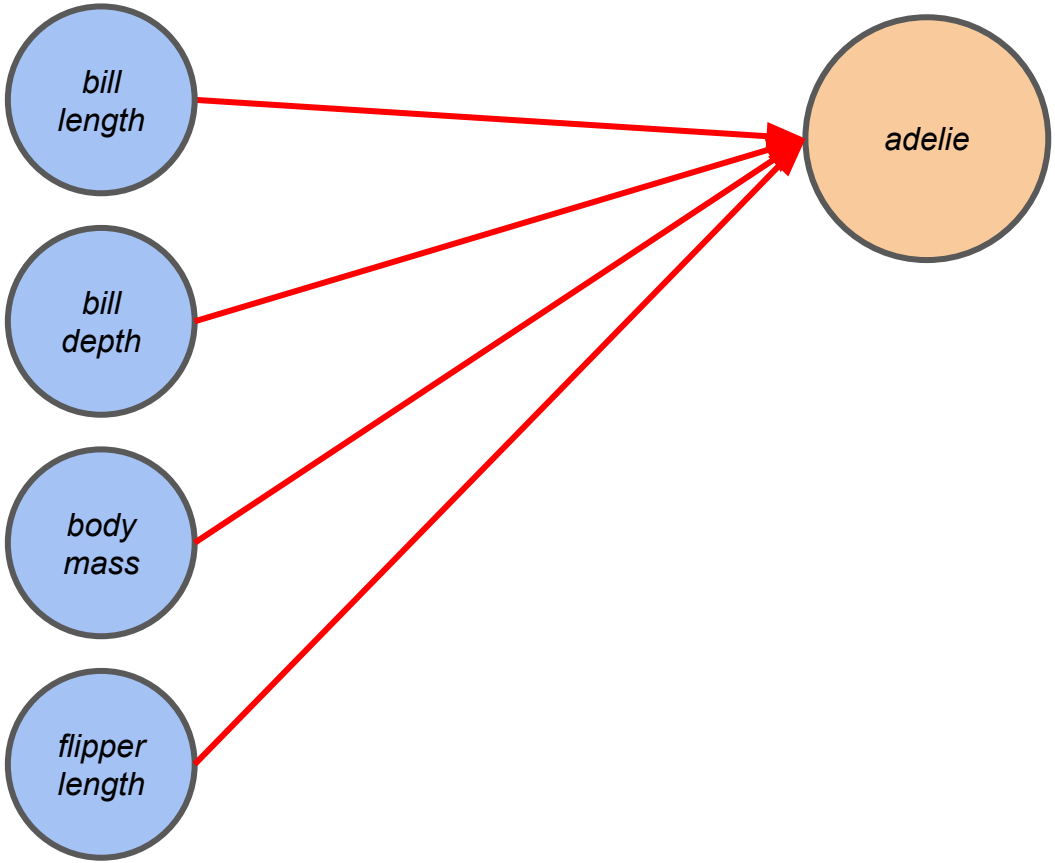
Neural Networks

Recommended Reading:

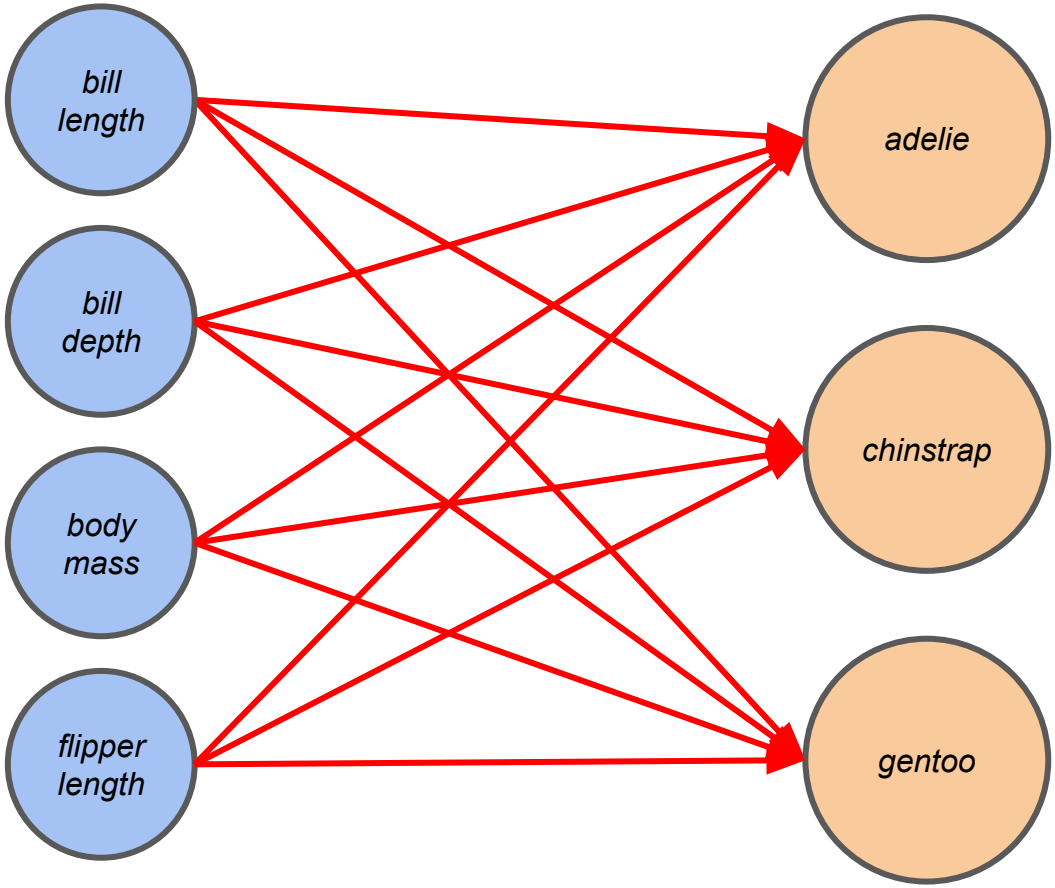
An Introduction to Statistical Learning, Sections 10.1 and 10.2

Download it for free here:
https://www.statlearning.com/





Logistic Regression:
Log odds of adelie is linear combination of the predictor features.



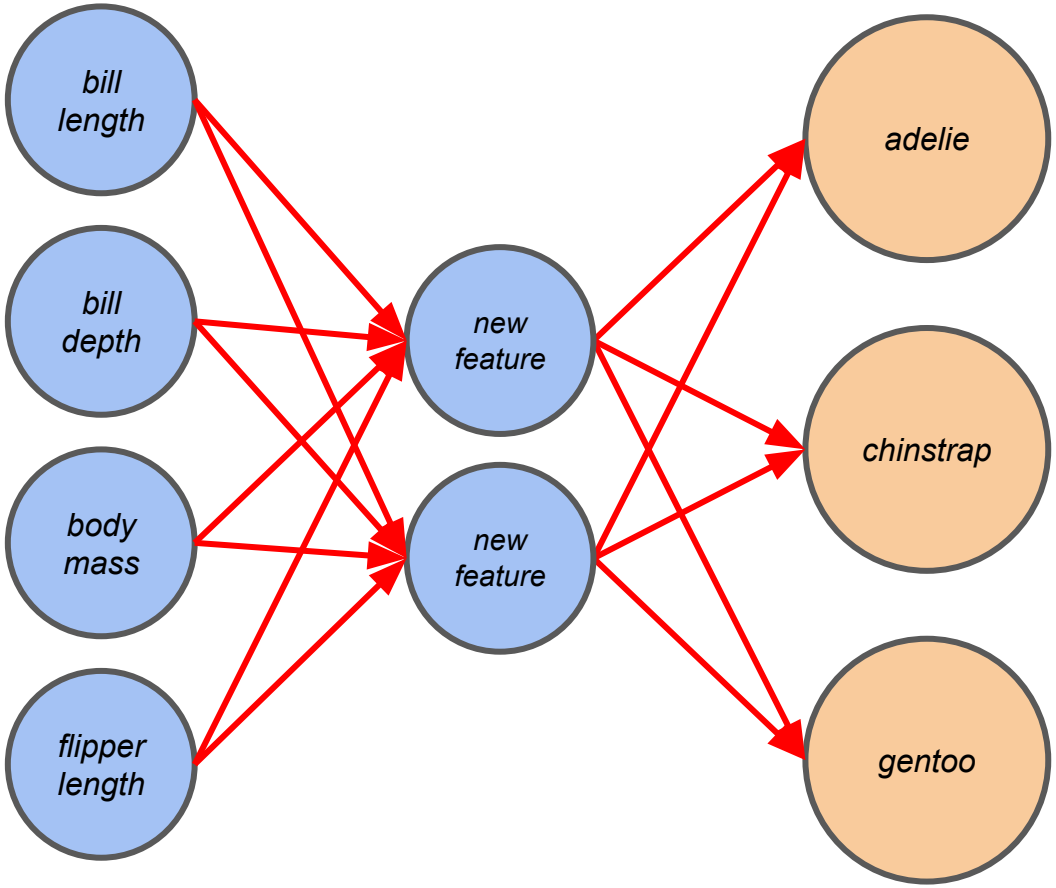
Logistic Regression:
Log odds of adelie is linear combination of the predictor features.

For multi-class, we have a separate set of coefficients for each.

We’ve see how we can create new features out of existing features through, for example, interaction terms or polynomial features.

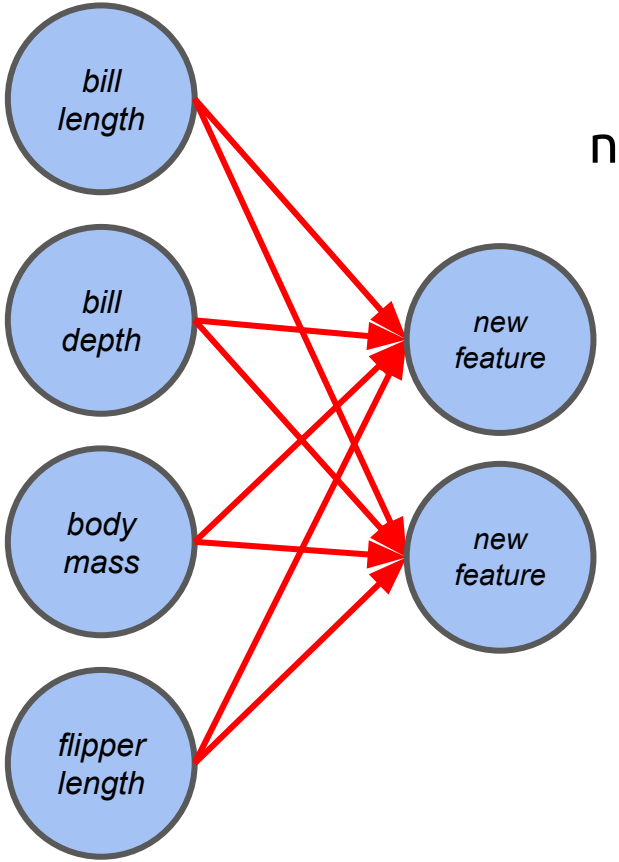
However, it requires some work to determine which features we want to include, or we have to spend time on hyperparameter tuning for models like ridge or lasso regression.

Another option is to create new features out of the input features automatically.



Neural networks: Create new features out of the starting features by taking linear combinations and passing through a (typically non-linear) **activation function**, like the hyperbolic tangent.

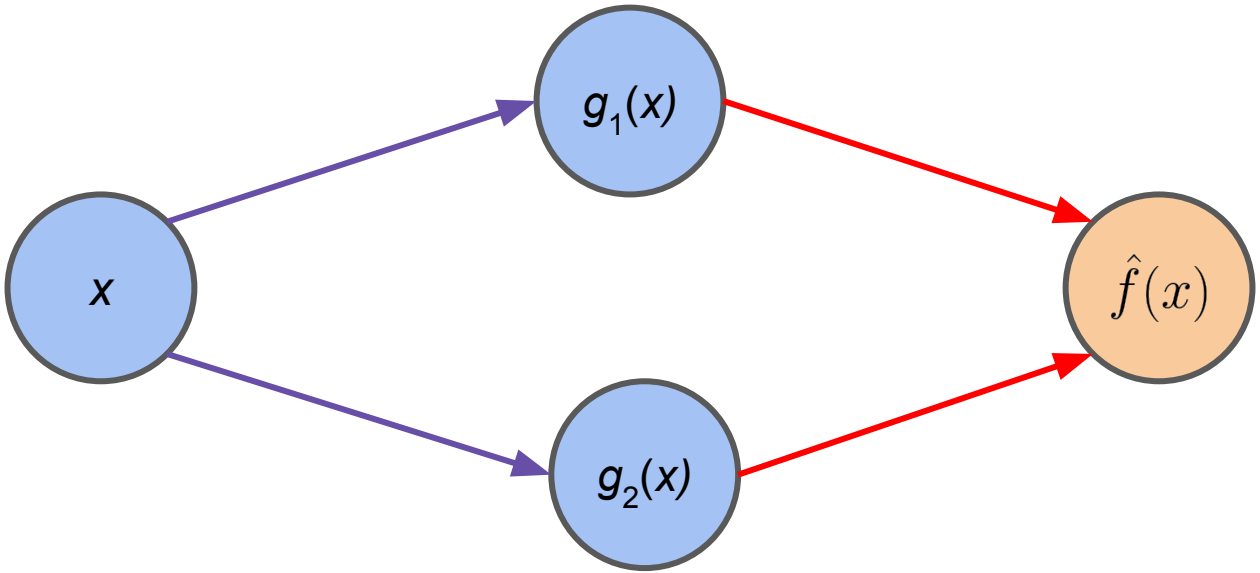
These new features are then fit to a logistic regression model (or linear regression, for numeric targets).



$$\text{new feature} = \tanh(a_0 + a_1(\text{bill length}) + a_2(\text{bill depth}) + a_3(\text{body mass}) + a_4(\text{flipper length}))$$

Terminology:

- The model we used is called a **neural network**.
- Each node is a **neuron**.
- The nonlinear function we used in each neuron (*tanh*) is called an **activation function.**



Other Parameters

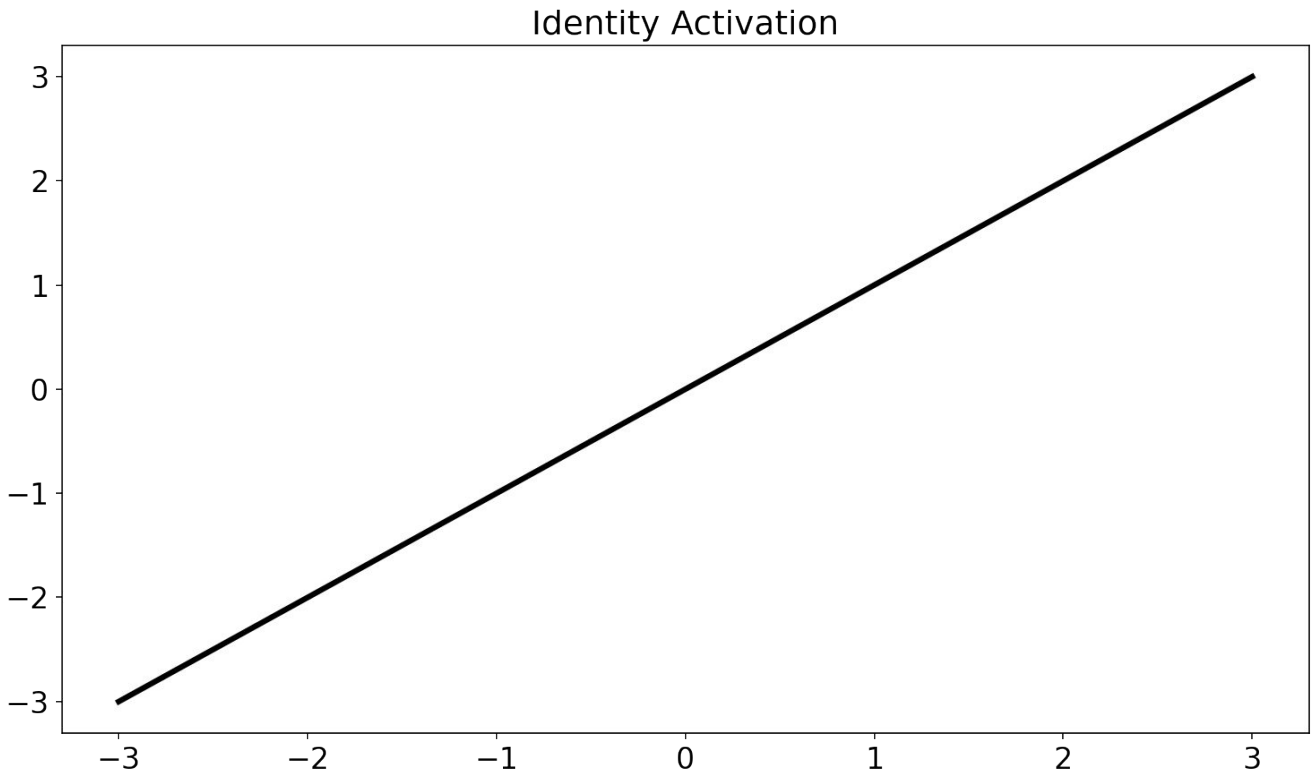
activation : *{'identity', 'logistic', 'tanh', 'relu'}, default='relu'*

Activation function for the hidden layer.

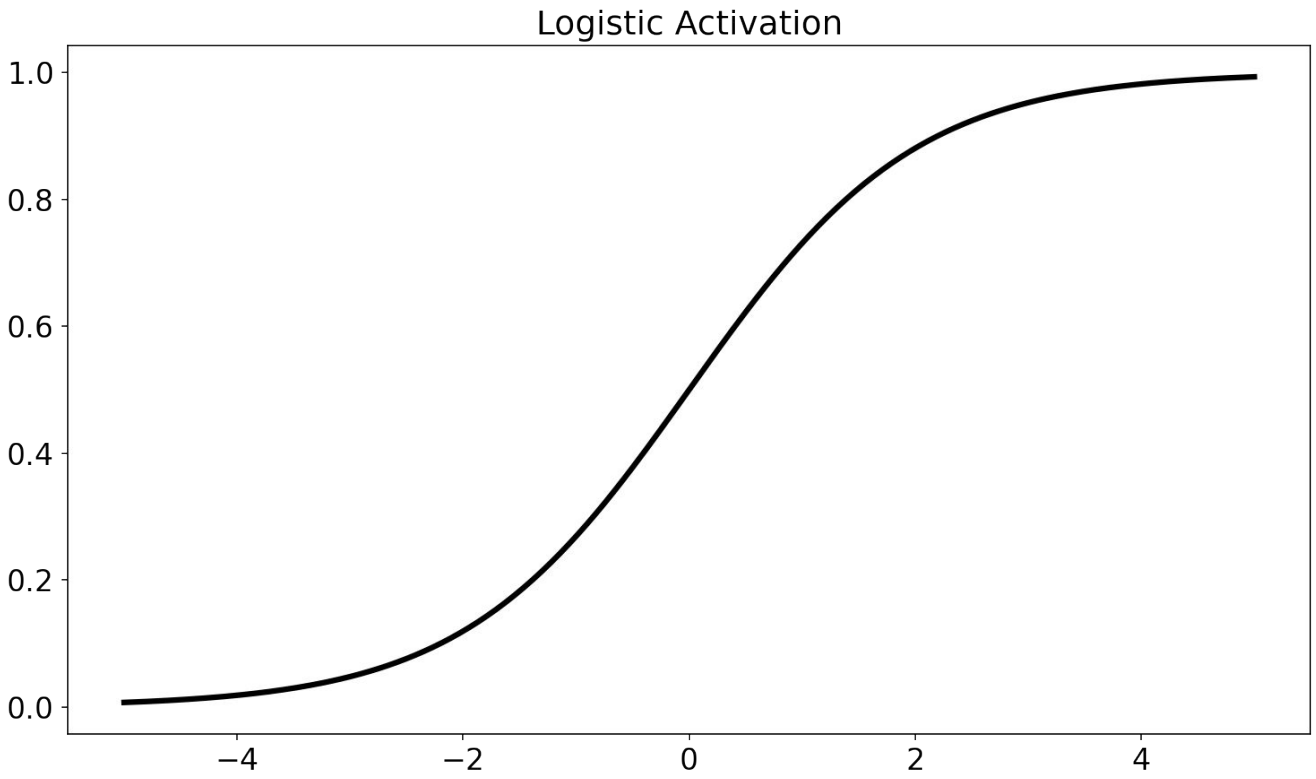
- 'identity', no-op activation, useful to implement linear bottleneck, returns f(x) = x
- 'logistic', the logistic sigmoid function, returns f(x) = 1 / (1 + exp(-x)).
- 'tanh', the hyperbolic tan function, returns f(x) = tanh(x).
- 'relu', the rectified linear unit function, returns f(x) = max(0, x)

The scikit-learn MLP classes give 4 options for activation functions.

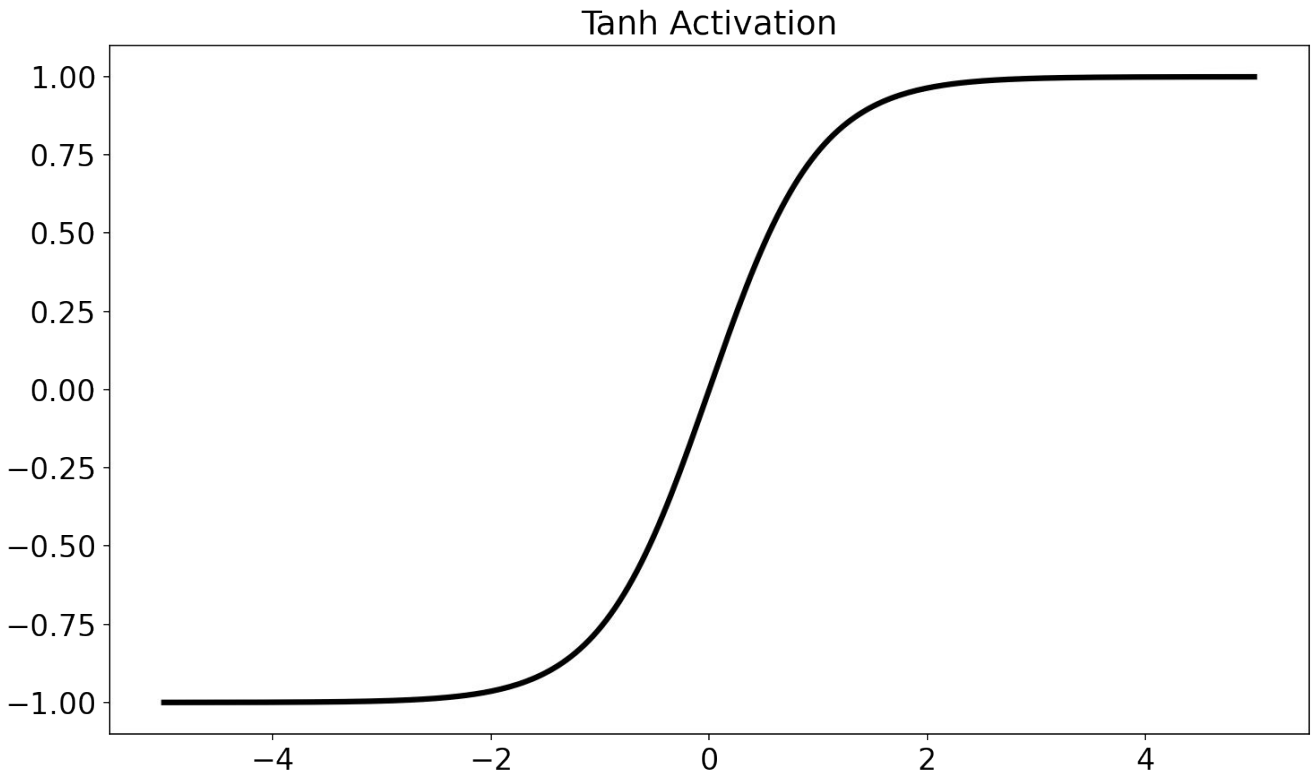
The identity activation does not give the benefits of a non-linear model.



This is the same function used for logistic regression.

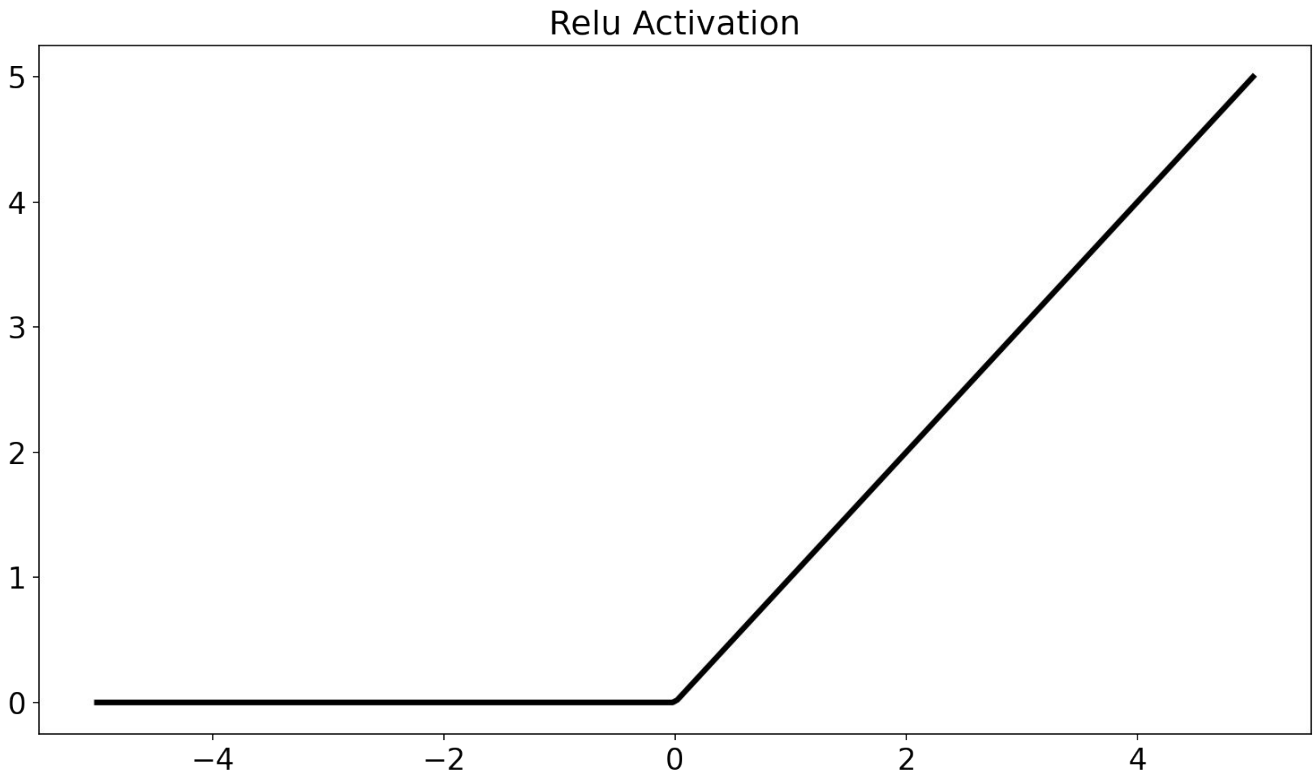


This activation is a bit steeper in the middle compared to logistic.



The rectified linear activation is a popular one.

$$\text{Relu}(x) = \max(0, x)$$



Other Parameters

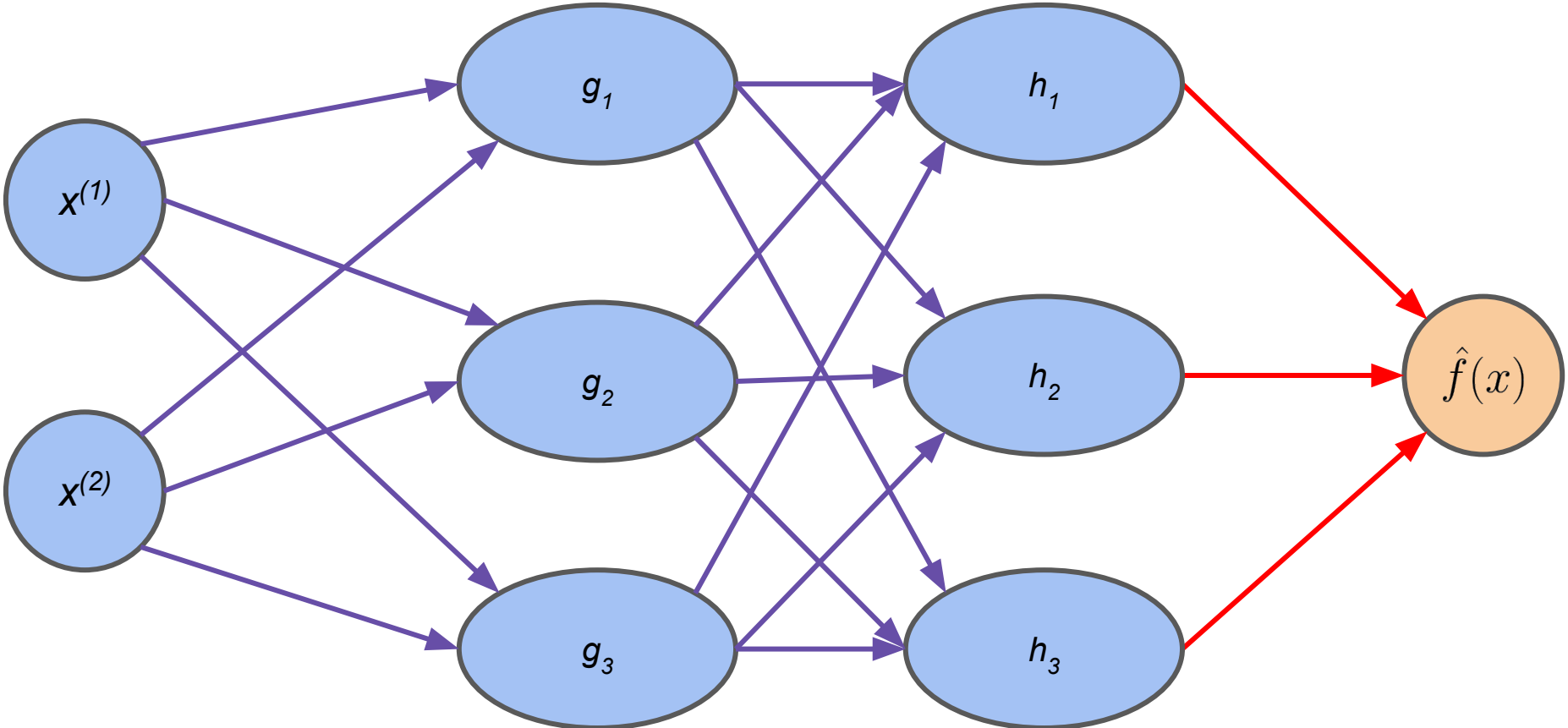
hidden_layer_sizes : *tuple, length = n_layers - 2, default=(100,)*

The ith element represents the number of neurons in the ith hidden layer.

You can specify how many layers you want and how many neurons per layer as a tuple.

By default, it will build a 100-dimensional representation of your data.

We can also do a few **hidden layers** of new features.
Deep learning is when there are more than one hidden layer.



Other Notes

When fitting a neural network, you will want to scale your features, either with a StandardScaler or MinMaxScaler. This helps the network converge faster.

For now, we'll use the MLPRegressor/MLPClassifier classes from scikit-learn, but later we'll see how to use Keras to more complex neural networks.